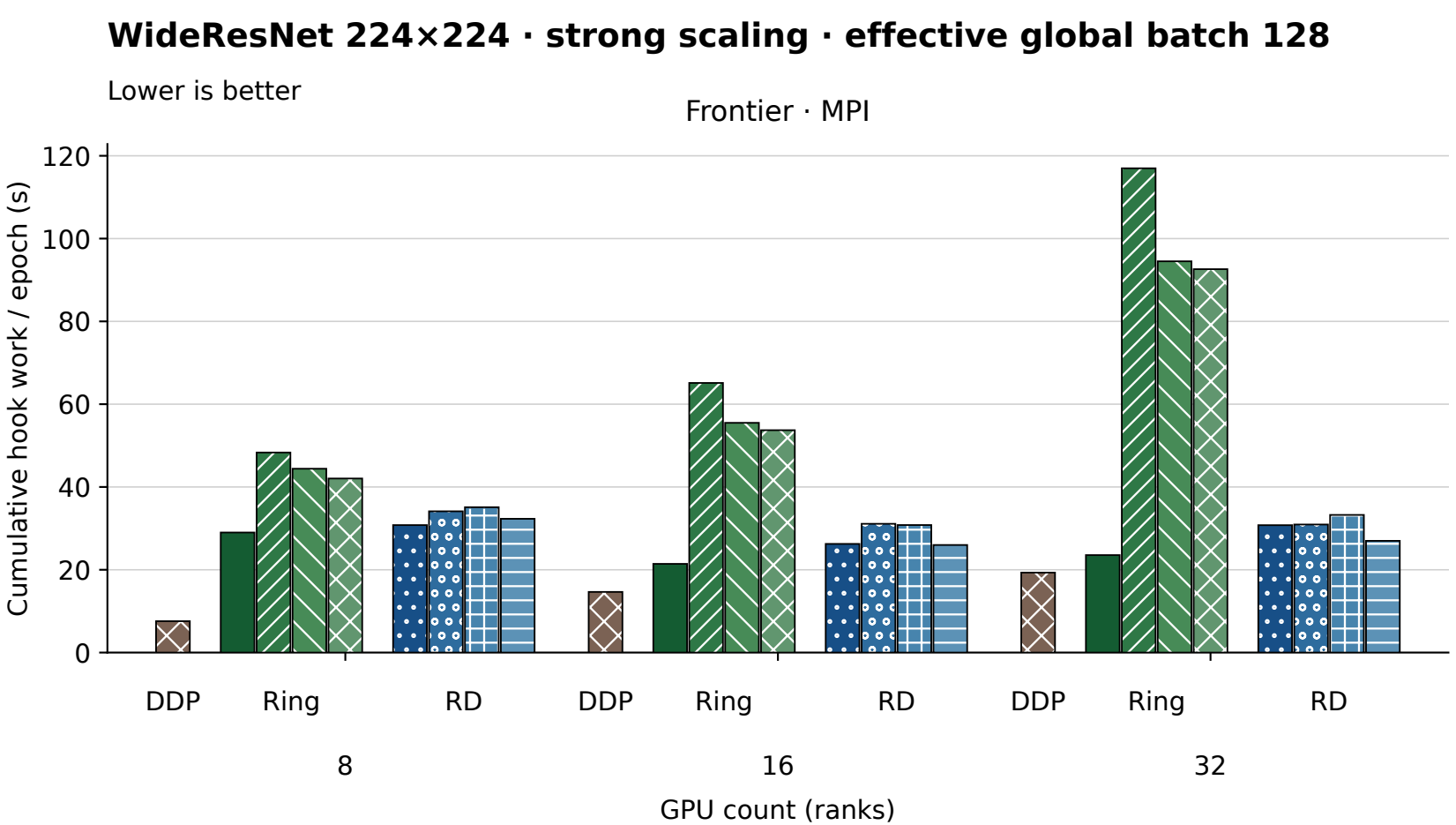

WideResNet 224×224 · strong scaling · effective global batch 128
Lower is better
Frontier · MPI
Cumulative hook work / epoch (s)
120
100
80
60
40
20
0
DDP
Ring
RD
DDP
Ring
RD
DDP
Ring
RD
8
16
32
GPU count (ranks)
DDP diagnostics
default sync
Ring
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Recursive doubling
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate